Separation of Rock Salt

Aim: To obtain pure salt from rock salt.

Equipment:

- Sample of rock salt
- Pestle and mortar
- Two 250cm^3 beakers
- Stirring rod
- Filter funnel
- Filter papers
- Tripod
- Heatproof mat
- Bunsen burner
- Evaporating basin

Safety Information

Hot water – risk of scalds and burns

Wear goggles, take care when heating

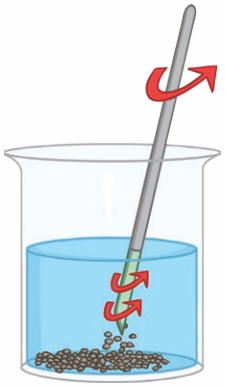
Glassware – risk of cuts if broken

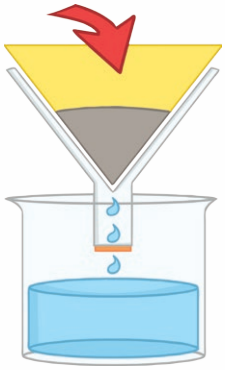
Apply pressure to wounds and inform teacher immediately

Bunsen burner – risk of burns

Place under cold running water for 10 minutes and inform teacher









Method:

Step 1: Carefully grind up the rock salt in the pestle and mortar.

Step 2: Stir the ground rock salt into a beaker of water to dissolve it.

Step 3: Carefully filter the solution through the filter paper in the funnel, into the second beaker.

Step 4: Transfer approximately 10ml of the filtered solution into the evaporating basin and set on the tripod over the Bunsen burner.

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Step 5: Heat the water until it is gently boiling and the water begins to evaporate off.

Step 6: When crystals begin to form on the sides of the basin, turn off the Bunsen burner – you now have a saturated solution.

Step 7: Leave the equipment to cool and then move the basin to a warm place to evaporate more slowly (overnight).



Rock salt is a mixture of sand and salt. It is also sometimes called grit salt and is used on roads during winter. The sand and salt compounds have different properties and so they can be separated.

Questions:

1. In step 2, water is acting as a solvent. What is a solvent?
2. Explain how the salt water solution is separated from the sand in step 3.
3. Draw a clearly labelled diagram to show the particles of sand and salt in step 3.
4. What is meant by a saturated solution, as described in step 6?
5. Explain, in terms of energy and particle arrangement, what evaporation is.

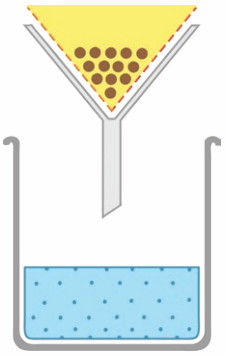
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Answers:

1. In step 2, water is acting as a solvent. What is a solvent?
   A solvent is a substance able to dissolve other substances.

2. Explain how the salt water solution is separated from the sand in step 3.
   The sand grains are insoluble (unable to dissolve) and so they cannot pass through the porous filter paper with the water and are left in the filter funnel.

3. Draw a clearly labelled diagram to show the particles of sand and salt in step 3.



4. What is meant by a saturated solution, as described in step 6?
   As much salt as possible is dissolved - no more salt can be dissolved.

5. Explain, in terms of energy and particle arrangement, what evaporation is.
   The water particles have gained more energy from the heating, and begin to move faster. The bonds between the water particles become weaker and some break off and become a gas particle instead.